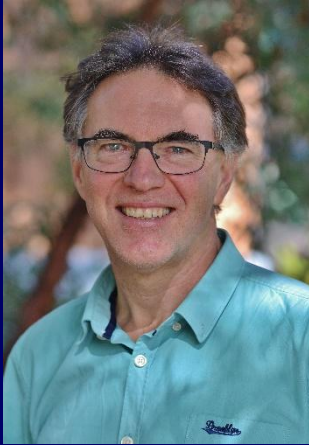


M17 – Zaragoza, Spain
22/06/2026-26/06/2026

***Equivariant Systems Theory and
Observer Design for Autonomous Systems***



Robert Mahony

Systems Theory and Robotics group
The Australian National University

Robert.Mahony@anu.edu.au

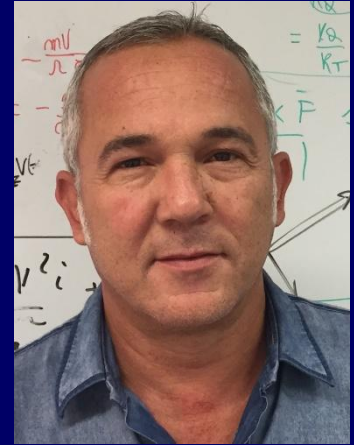
<http://users.cecs.anu.edu.au/~Robert.Mahony/>



Jochen Trumpf

Systems Theory and Robotics group
The Australian National University

jochen.trumpf@anu.edu.au



Tarek Hamel

Laboratoire d'Informatique de Signaux
et Système de Sophia-Antipolis (I3S),
CNRS-UNSA, France

thamel@i3s.unice.fr

<http://www.i3s.unice.fr/~thamel/>

Abstract of the course

The functionality of any autonomous system depends critically on its ability to estimate its dynamic state. For robotic systems, with limited sensor suites, highly dynamic motion, non-linear state space, and limited computational capacity, the state observer performance is even more important. A key technology enabler underlying the explosion of small scale commercial aerial robotic systems was the development of high quality, simple, robust, attitude observers based on the underlying equivariant symmetry structure of the attitude estimation problem.

This course provides an introduction to emerging field of *Equivariant Systems Theory* and applies these techniques to the design of observers for autonomous systems. The approach taken is highly practical, starting from matrix calculus and Lie theoretic foundations from an engineering perspective and working through a number of examples to demonstrate how to model equivariant systems and use the symmetry to derive robust observers. The course is based around an extensive suite of case studies drawn from aerial robotic applications including; attitude estimation, velocity aided attitude estimation, pose estimation, homography estimation, SLAM and Visual Odometry. Students will come out of the course with a strong understanding of how to derive and implement equivariant observers and filters for real world robotic systems.

Topics:

- 1) Perspectives on observer design for autonomous systems.
- 2) Matrix calculus and matrix ODEs.
- 3) Lyapunov observer design for systems with matrix Lie-group state.
- 4) Dealing with practical issues, velocity bias, asynchronous measurements and delays.
- 5) Equivariant Systems Theory.

Practical work in this course uses MATLAB (or equivalent scripting language such as Python) extensively. Students are required to have a working system on their own laptop for the course.